

Data Sheet Miniature Temperature Controller MTD415T



Features

- Small, safe and very high accuracy complete single-module controller
- High speed, ultra stable digital PID Loop
- Up to ± 1.5 A TEC current
- Very low output current noise
- Very small footprint (21.0 x 12.4 mm)
- Circuit Height 3.1 mm
- Supports 10 k Ω thermistor temperature sensor
- Single power supply operation

Applications

- Active cooling and temperature stabilization for a wide range of laser modules and diodes
- WDM, DWDM Laser-Diode Temperature Control
- EDFA Optical Amplifiers
- Temperature stabilization of photo detectors and photodiodes
- ATE

Short Description and Typical Application Diagram

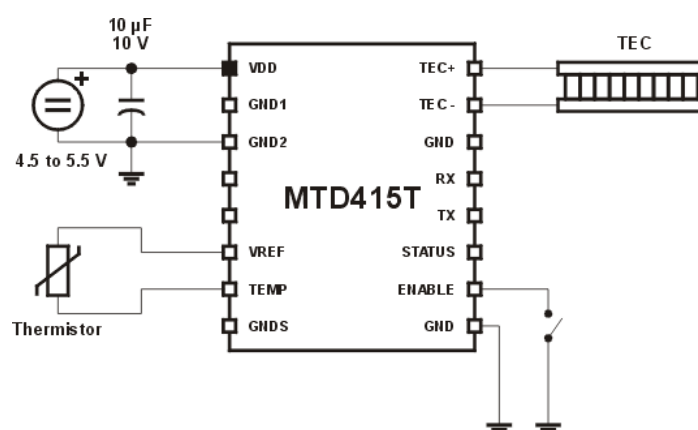
The MTD415T is a compact and highly integrated temperature controller optimized for use in high performance thermoelectric temperature control applications.

The on-chip power stage and the thermal control loop circuitry minimize external components while maintaining high efficiency.

The output current is directly controlled to eliminate current surges. An adjustable TEC current limit provides the highest level of TEC protection.

The MTD415T is operated from a single power supply and provides a bipolar ± 1.5 A output by connecting the TEC to the output of a bipolar power stage. True bipolar operation ensures temperature control without “dead zones” or other nonlinearities at low TEC current values.

The digital control interface allows quick access to all system parameters as well as to digital measurement data, this way enabling a simple integration into different systems.



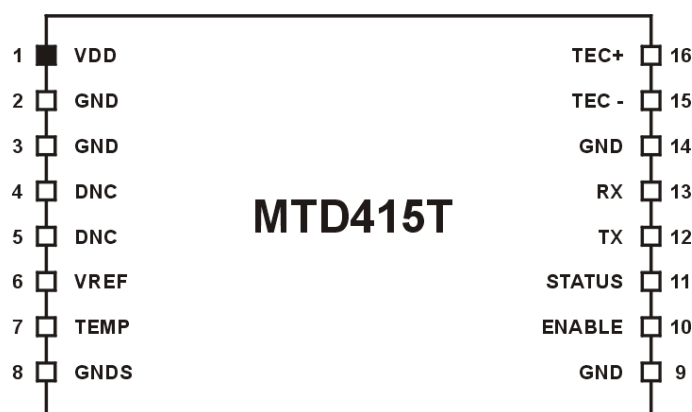
Revision History

Revision	Changes with respect to previous revision
1.0	Initial Release

Contents

1	Pin Configuration and Functions	4
2	Technical Data	6
2.1	Absolute Maximum Ratings	6
2.2	Recommended Operating Conditions	6
2.3	Electrical Characteristics	7
3	Typical Output Characteristics	8
4	Functional Block Diagram	9
5	Typical Application	11
6	Programmers Reference	12
6.1	Nomenclature	12
6.2	Command Description	12
6.2.1	General Commands	12
6.2.2	TEC Commands	13
6.2.3	Temperature	14
6.2.4	Control Loop	15
6.2.5	Save Settings	17
6.2.6	Factory Default Settings	17
6.3	Error Register and Safety Bitmask	18
7	PID Tutorial	20
8	Troubleshooting	22
9	Drawing	23
10	List of Acronyms	24
11	Warranty	25
12	Copyright and Exclusion of Reliability	26
13	Thorlabs 'End of Life' Policy	27
14	Thorlabs Worldwide Contacts	28

1 Pin Configuration and Functions



MTD415T Pin Configuration

Pin	Name	Description
1	VDD	Supply Voltage Input Connect a + 4.5 V to 5.5 V power supply to VDD.
2	GND	Supply Voltage Ground Connect the power supply ground connections to GND.
3	GND	Supply Voltage Ground Connect the power supply ground connections to GND.
4	DNC	Do Not Connect Do not connect this pin to any signal or potential. This pin is used for manufacturing and test purposes.
5	DNC	Do Not Connect Do not connect this pin to any signal or potential. This pin is used for manufacturing and test purposes.
6	VREF	Reference Output Voltage for Thermistor Temperature Sensor Connect this reference voltage output (1.8 V) to one end of the 10 kΩ thermistor.
7	TEMP	Thermistor Temperature Sensor Input Connect this pin to the other end of the 10 kΩ thermistor.
8	GNDS	Temperature Sensor Ground Can be used for shielding purposes or left open.
9	GND	Supply Voltage Ground Connect the power supply ground connections to GND.
10	ENABLE	Enable Signal Input (Low-Active) Enable Input (Low = enabled, High = Disabled), can be connected directly to GND.
11	STATUS	Status Signal Output (Can be left floating) Status Signal (High = temperature within defined temperature window, Low = Temperature outside programmed temperature window or an error occurred).
12	TX	Digital Interface Transmit Signal UART Transmit Asynchronous Data Output. Connect this pin to the RX pin of your application.

Pin	Name	Description
13	RX	Digital Interface Receive Signal UART Transmit Asynchronous Data Input. Connect this pin to the TX pin of your application.
14	GND	Supply Voltage Ground Connect the power supply ground connections to GND.
15	TEC -	TEC Element negative connection Connect this pin to the negative terminal of the TEC element.
16	TEC +	TEC Element positive connection Connect this pin to the positive terminal of the TEC element.

2 Technical Data

2.1 Absolute Maximum Ratings

Supply Input Voltage	4.5 V to 6 V
Supply Input Current	1.6 A
TEC Output Current	-1.5 A to 1.5 A
TEC Compliance Voltage	4.0 V
Maximum Output Power	6.0 W
Power Dissipation	1.5 W
Pin Voltage Range ¹⁾ VDD ENABLE, RX, TEC-, TEC+ TEMP	-0.3 V to 6 V -0.3 V to (VDD + 0.3 V) -0.3 V to 3.3 V
Maximum Output Current STATUS, TX	10 mA
Maximum Input Current ENABLE, RX	10 mA
Operating Temperature	-40 °C to + 70 °C

¹⁾ All voltages with respect to network ground terminal.

Notes

- (1) Above specifications are given for the free-air operating temperature range unless otherwise noted.
- (2) Stresses beyond those listed above may cause permanent damage to the product. These are stress ratings only; functional operation of the MTD415T at these or any other conditions beyond those indicated under [Recommended Operating Conditions](#)⁶⁾ and [Electrical Characteristics](#)⁷⁾ is not implied.
- (3) Operation beyond the maximum rated conditions for extended periods may affect product reliability.

2.2 Recommended Operating Conditions

Supply Voltage	4.5 to 5.5 V
Operating Temperature	-20 to + 60 °C

2.3 Electrical Characteristics

TEC Current Output	
Output Current	up to ± 1.5 A into 2.66Ω TEC - see diagram on page 8
Compliance Voltage	4.0 V
Output Power	up to 6.0 W
Measurement Resolution	better than 8 mA, typ. 3 mA
Measurement Accuracy	± 50 mA
Noise and Ripple (typ.)	< 10 mA ¹⁾
TEC Current Limit	
Setting Range	0 to 1.5 A
Setting Resolution	1 mA
Setting Accuracy	± 50 mA
Temperature Sensor	
Supported Sensor	10 k Ω Thermistor ⁵⁾
Maximum Temperature Control Range ²⁾	+ 5 °C to + 45 °C
Temperature Setting Resolution	1 mK
Temperature Measurement Resolution ³⁾	better than 10 mK; typ. 2 mK
Absolute Temperature Accuracy ²⁾	± 0.5 °C
Temperature Stability over 8 h, typ.	better than 20 mK
Temperature Coefficient	< 20 mK/°C
Programming Interface	
Type	UART
Voltage Level	3.3 V Logic Level; input 5 V tolerant
Data Rate	115.200 bps; 8 Data Bits, 1 Stop Bit
General Data	
Safety Features	• TEC Current Limit • Sensor Fault Protection • TEC Open Circuit Protection • Temperature Setpoint Limit • Temperature Window Protection Delay • Over Temperature Protection
Operating Temperature	-20 °C to +60 °C ⁴⁾
Storage Temperature	-40 °C to +100 °C ⁴⁾
Warm-Up Time for Rated Accuracy	10 min
Dimensions (W x H x D)	21 x 12.4 x 3.1 mm ³
Approx. Weight	2 g

¹⁾ Measured with a TEC element with an equivalent resistance of 4 Ω .

²⁾ Control range and thermal stability depend on thermistor parameters.

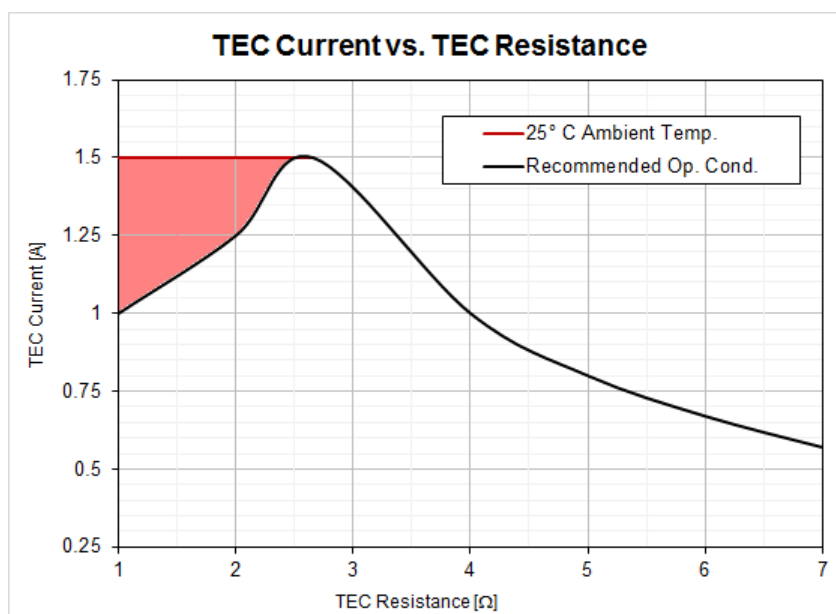
³⁾ Maximum measurement resolution depends on cycle time settings. Please refer to Programmers Reference.

⁴⁾ non-condensing

⁵⁾ Use only thermistors with 10k Ω resistance at 25 °C ($R_{25^{\circ}\text{C}}$). Resistance Measurement Range: 4.2 to 29 k Ω

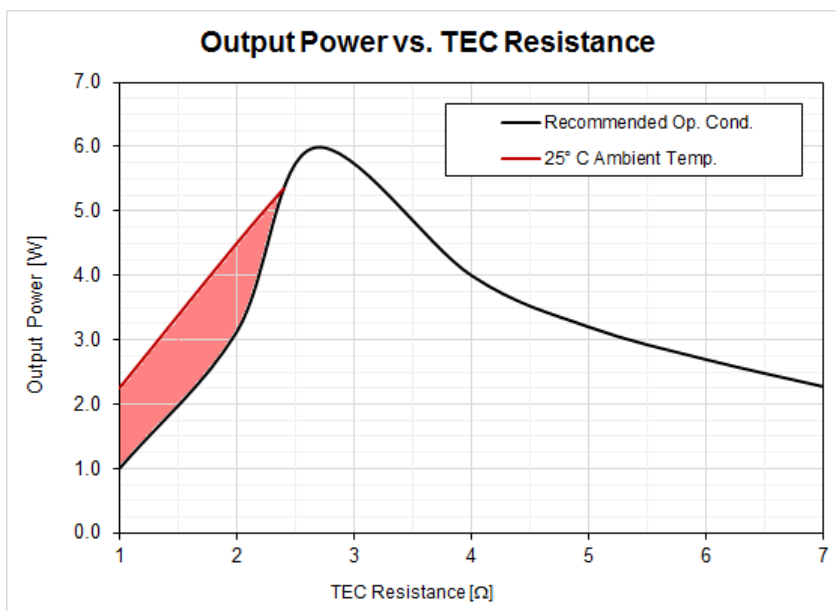
All technical data are valid at 23 ± 5 °C and $45 \pm 15\%$ rel. humidity (non condensing)

3 Typical Output Characteristics



Notes

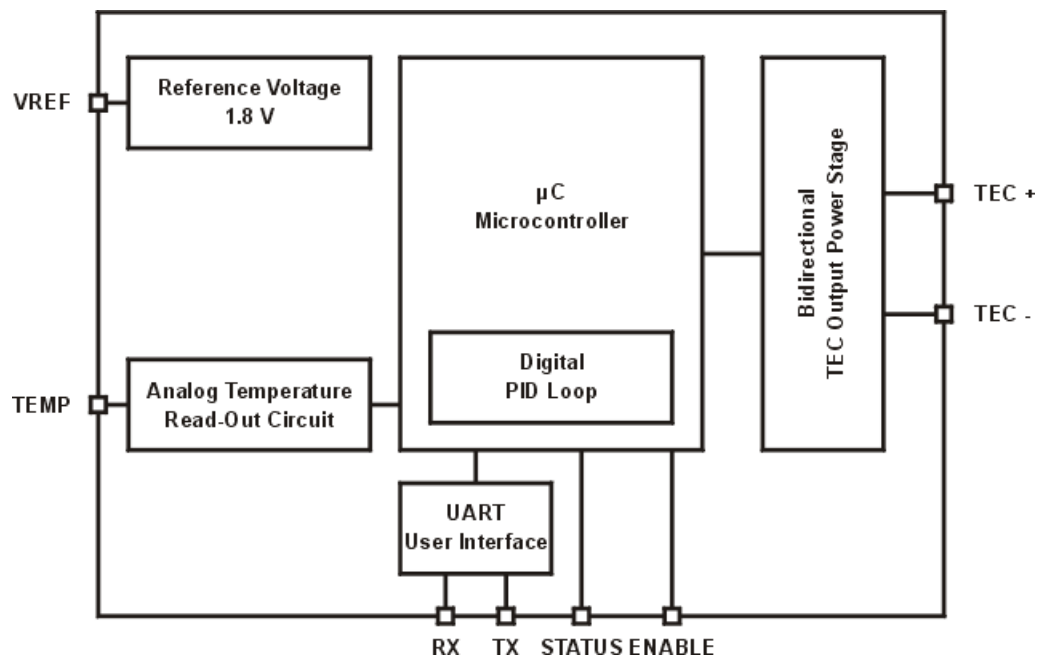
1. The maximum TEC current of 1.5 A can be delivered into a load resistance of 2.66 Ω at recommended operating conditions.
2. At higher load resistance, the maximum output current drops due to compliance voltage limitation.
3. The maximum output current at lower than 2.66 Ω load resistance (colored range) depends on environmental conditions.



Notes

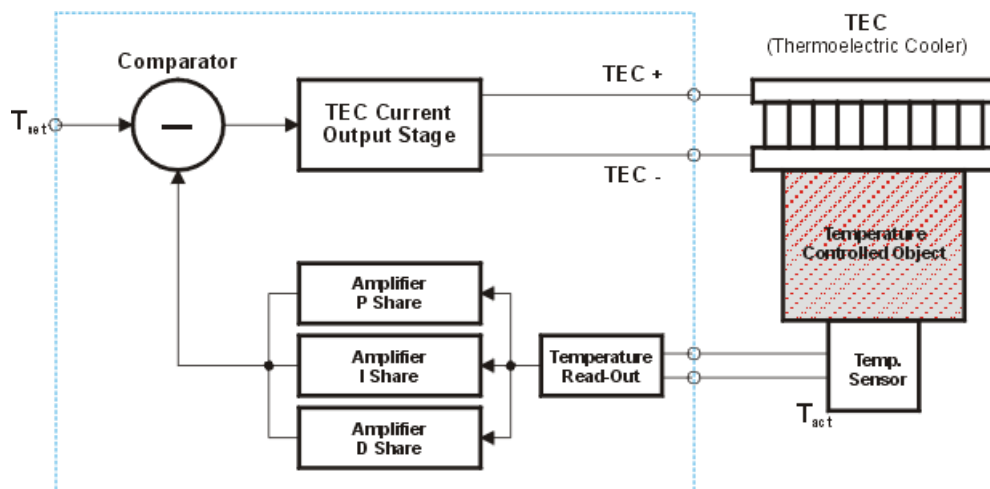
1. The maximum output power of 6 W can be delivered into a load resistance of 2.66 Ω at recommended operating conditions.
2. The maximum output power at lower than 2.66 Ω load resistance (colored range) depends on environmental conditions.

4 Functional Block Diagram



Functional Block Diagram MTD415T

Principle of Operation



In general, a temperature controller (within the blue frame) is a closed loop system. A temperature sensor measures the temperature of the controlled object (e.g., a laser diode). This **actual temperature** signal is amplified and compared with the **temperature set value**. The differential signal out of the **comparator** controls then the current of the **thermoelectric cooler** in order to maintain the temperature of the object constant. Ideally, the temperature settling is carried out in the shortest times, with minimum settling error and without temperature overshoots.

A thermoelectric cooler is a Peltier element that produces a temperature gradient depending on the current direction through the TEC. For this reason, the TEC current must be bidirectional. In order to adapt the control loop to different thermal loads, and to optimize the temperature controller's response characteristics, a PID amplifier is used. Closer information please see in the PID Tutorial ([page 20](#))

Functional Description

The MTD415T is a miniature, closed-loop temperature controller module. It is compatible with a 10 k Ω NTC (thermistor) temperature sensor and its output is designed for control of thermoelectric coolers (TEC).

The MTD415T delivers a TEC current up to 1500 mA at 4.0 V compliance voltage.

Power Supply

The supply voltage ranges from 4.5 V to 5.5 V. From the supply voltage, the internal supply voltage for the microcontroller is derived. Further, a reference voltage for the temperature sensor is generated (**VREF**; 1.8 V).

TEC Current Control

The TEC element is connected between **TEC+** and **TEC-**. A correct connection is essential to avoid wrong temperature correction.

The MTD415T allows to limit the maximum TEC current. Lowering the TEC current limit might be helpful for control loop optimization in case of low thermal loads and lowers the dissipated by the MTD415T power.

Micro-controller

The functions of the microcontroller are:

- Comparison of the actual temperature with the set temperature;
- Generation of the control signal for TEC output stage with respect to the comparator signal;
- PID loop control for optimization of the temperature settling time and for minimizing the final temperature error.

All parameters (TEC current limit, set temperature, temperature window, PID share settings etc.) are programmed via the **UART** user interface. Detailed information about how to program the MTD415T can be found in the section Programmers Reference on [page 12](#).

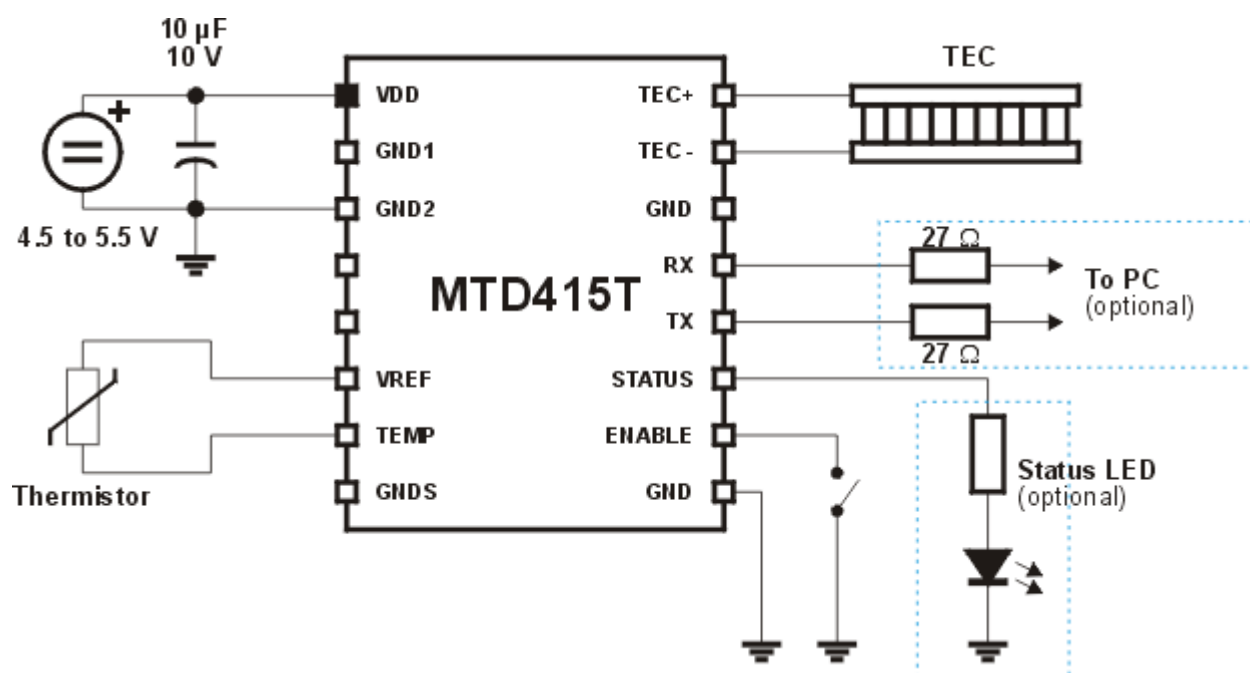
Note

The **UART** interface uses 3.3 V logic level. Connect the **UART** to a PC only using an appropriate converter, e.g., a commercially available UART-to-USB cable.

The **Status** signal informs about correct operation.

The temperature control is activated by setting the **Enable** pin to low.

5 Typical Application



6 Programmers Reference

6.1 Nomenclature

Program messages are written
in inverted commas:

"A?"

"Lx" (x - Parameter)

Response messages are written in brackets:

[-800 <LF>]

Data Format

Numerical value with sign in integer notation:
-220 or 16789432.

Command /response Terminator:

Line Feed (<LF>)

6.2 Command Description

6.2.1 General Commands

Command	Explanation Response Example
"m?"	Reads the version of hardware and software [MTD415T FW0.6.8]

Command	Explanation Response Example
"u?"	Reads the UUID (Universal Unique Identifier) of the MTD415T [045F778655FDE5118ED499C9B4521485]

Command	Explanation Response Example
"E?"	Reads the Error Register. Responses see section Error Register and Safety Bitmask ¹⁸
"c"	Resets the Error register

Note

The MTD415T has a non-volatile memory (flash) that stores the setting parameters. This memory has a limited number of erase / write cycles. In order to protect the flash memory, changes to setting parameters are not stored automatically. If you want to keep parameter changes after power-down of the MTD415T, save them to the flash memory using the "M" command. This command saves the T, W, L, d, G, O, P, I, D, C and S parameters at a time.

6.2.2 TEC Commands

Command	Explanation Response Example
Programming	
"Lx"	Sets the TEC current limit to x *) Value range x : 200 to 2000 [mA]
Reading	
"L?"	Reads the TEC current limit [x<LF>] [mA]
"A?"	Reads the actual TEC current [x<LF>] [mA] x < 0: Heating; x > 0: Cooling
"U?"	Reads the actual TEC voltage [x<LF>] [mV]

*) This set value remains during the actual power-on cycle. Use the "**M**" command to save it to the non-volatile memory.

6.2.3 Temperature

Command	Explanation Response Example
Programming	
"Tx"	Sets the set temperature to x *) Value range x : 5000 to 45000 [10 ⁻³ °C]
Reading	
"T?"	Reads the set temperature [x <LF>] Value range x : 5000 to 45000 [10 ⁻³ °C]
"Te?"	Reads the actual temperature [x <LF>]

Command	Explanation Response Example
Programming	
"Wx"	Sets the set temperature window to x *) Value range x : 1 to 32768 [mK]
Reading	
"W?"	Reads the temperature window [x <LF>] [mK]

Command	Explanation Response Example
Programming	
"dx"	Sets the delay time between reaching the temperature window and activating the Status output pin to x *) Value range x : 1 to 32768 [sec]
Reading	
"d?"	Reads the delay time between reaching the temperature window and activating the Status output pin [x <LF>] [sec]

*) This set value remains during the actual power-on cycle. Use the "**M**" command to save it to the non-volatile memory.

6.2.4 Control Loop

Loop Test

Command	Explanation Response Example
Programming	
"Gx"	Sets the critical gain to x *) Value range x: 10 to 100000 [mA/K]
Reading	
"G?"	Reads the critical gain [x<LF>] [mA/K]

Command	Explanation Response Example
Programming	
"Ox"	Sets the critical period to x *) Value range x: 100 to 100000 [msec]
Reading	
"O?"	Reads the critical period [x<LF>] [msec]

PID Settings

Command	Explanation Response Example
Programming	
"Cx"	Sets the cycling time to x *) Value range x: 1 to 1000 [msec]
Reading	
"C?"	Reads the cycling time [x<LF>] [msec]

Command	Explanation Response Example
Programming	
"Px"	Sets the P Share to x *) Value range x: 0 to 100000 [mA/K]
Reading	
"P?"	Reads the P Share [x<LF>] [mA/K]

*) This set value remains during the actual power-on cycle. Use the "M" command to save it to the non-volatile memory.

Command	Explanation Response Example
Programming	
"I $\mathbf{x}$ "	Sets the I Share to $\mathbf{x}$ *) Value range $\mathbf{x}$: 0 to 100000 [mA/(K+sec)]
Reading	
"I?"	Reads the I Share [$\mathbf{x}$ <LF>] [mA/(K+sec)]

Command	Explanation Response Example
Programming	
"D $\mathbf{x}$ "	Sets the D Share to $\mathbf{x}$ *) Value range $\mathbf{x}$: 0 to 100000 [(mA*s)/K]
Reading	
"D?"	Reads the D Share [$\mathbf{x}$ <LF>] [(mA*sec)/K]

*) This set value remains during the actual power-on cycle. Use the "**M**" command to save it to the non-volatile memory.

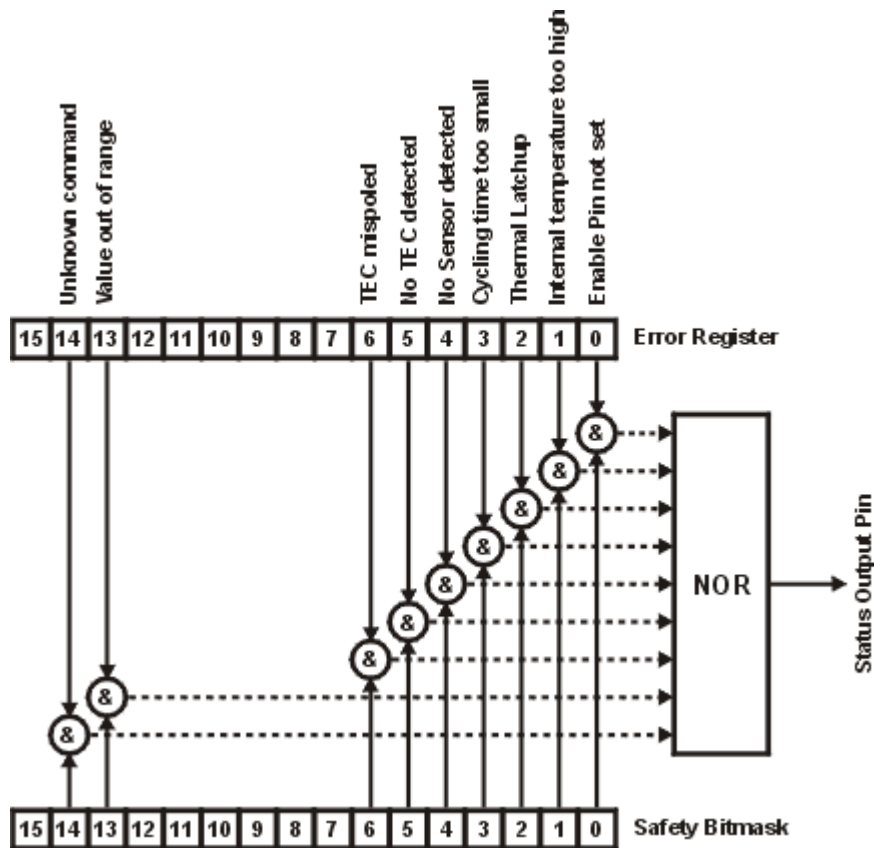
6.2.5 Save Settings

Command	Explanation Response Example
"M"	Saves the setup. The actual parameters that have been set using the commands T, W, L, d, G, O, P, I, D, C and S, are saved to the nonvolatile memory.

6.2.6 Factory Default Settings

Parameter	Explanation	Factory Default
L	TEC current limit	1000 mA
T	Temperature set value	25 °C
W	Temperature window	1000 mK
d	Temperature window delay	10 sec
C	Cycle time PID loop	50 msec
P	P share PID loop	1000 mA/K
I	I share PID loop	200 mA/(K*sec)
D	D share PID loop	100 (mA*sec)/K
G	Critical PID loop gain (Loop test)	2000 mA/K
O	Critical PID loop oscillation period (Loop test)	2000 msec
S	Value of the Safety Bitmask	255

6.3 Error Register and Safety Bitmask



The MTD415T has an internal 16 bit error register:

Bit Number	Event
0	Enable pin not set to L (GND)
1	Internal temperature too high
2	Thermal Latch-Up (TEC current at limit without temperature improvement)
3	Cycling time too small
4	No Sensor detected
5	No TEC detected (connection open)
6	TEC mispoled
7	(not used)
8	(not used)
9	(not used)
10	(not used)
11	(not used)
12	(not used)
13	Value out of range
14	Invalid command
15	(not used)

The error register can be read out using the "**E?**" command. The error register can be reset using the "**c**" command or by setting the **Enable** pin to Off and On again.

Further, with default setting of the Safety Bitmask (value = 255), the states of bits 0 to 7 are handed over to the **Status** output pin - if any of the bits 0 to 7 is H, the **Status** level is "LOW" and the TEC Current Output is switched off.

The Safety Bitmask can be used to mask any desired error at the **Status** output pin.

Attention

Masking an error may lead to damage of the MTD415T. For example, if bit 1 is masked, exceeding the internal temperature will **not be reflected** on the status pin as an error and the TEC output will not be disabled!

The safety bitmask can be programmed:

Command	Explanation Response Example
Programming	
"Sx"	Sets the Safety Bitmask to x Value range x : 0 to 32768
Reading	
"S?"	Reads the Safety Bitmask value. [x <LF>]

Note

If the safety bitmask should be saved for future use, it needs to be memorized using the **"M"** command.

7 PID Tutorial

The general requirements to a temperature control loop are:

- fastest settling time after power on or changing the set temperature
- minimum residual temperature error
- settling without temperature overshoots
- fastest response to changes of the thermal load

PID amplifiers can fulfill these requirements. Temperature control loops are comparatively slow; control oscillations appear with a frequency in the range of several Hz or parts of Hz. The PID adjustment allows to optimize the dynamic behavior.

The **P share** is the proportional share, or the gain of the amplifier, that defines the settling time. The higher the P share, the faster the settling and the less residual temperature error. The downside is that high P shares lead to oscillations.

The **I share** is the integrating share of the amplification, or the gain at low frequencies. It allows to minimize the residual temperature error.

Optimal settings of the P and I shares result in a fast approach to the set temperature, without oscillations and with a minimum residual temperature error. However, such a loop is not able to quickly react to sudden changes of the thermal load, for example, if a thermally stabilized laser diode is set to a higher or lower output power that changes the laser's heat dissipation. The **D share** (differential share, or the gain at high frequencies) allows the system to quickly react to temperature changes, without generating oscillation of the temperature around the set point.

The MTD415T microcontroller incorporates a digital PID controller. The P, I and D shares can be programmed manually or calculated automatically by the firmware by entering the results of a loop oscillation test. Below an example procedure is explained in detail.

Example of a PID adjustment

Pre-conditions:

- TEC current limit is set correctly
- all connections are made properly

In order to observe the temperature change, connect an appropriate instrument that allows to display the temperature change vs. time, to the **TEMP** input of the MTD415T.

1. Configure the PID loop:

Set temperature = 25°C: "**T25000**"

P share = 1000 mA/K: "**P1000**"

I share = 0: "**I0**"

D share = 0: "**D0**"

Cycle time = 30 ms: "**C30**"

2. Enable the TEC. The actual temperature **T_e** approximates the set value.

3. Now, find the critical P share (critical gain) value at which the system starts to oscillate for a minimum of 20 cycles without amplitude drop as a reaction to a changed set temperature.

An example procedure is described below:

- Set P to 10.000 mA/K: "**P10000**".
In order to trigger loop oscillation, increase the set temperature for 0.1 K: "**T25100**"
- Lower P to 5.000 mA/K: "**P5000**", decrease the set temperature for 0.1 K: "**T25000**" and observe the loop behavior.
- If the loop still oscillates, lower the P share again, change the set temperature for 0.1K and observe the loop behavior.

Example:

"P3000"; "T25100" -> the loop still oscillates

"P2000"; "T25000" -> the loop stopped oscillating

- If the loop stopped oscillating, increase the P share, change the set set temperature for 0.1K and observe the loop behavior.

Example:

"P2500"; "T25100" -> the loop oscillates again

- Repeat above steps.

Example:

"P2200"; "T25100" -> the loop stopped oscillating.

"P2300"; "T25000" -> the loop oscillates again

"P2250"; "T25100" -> the loop still oscillates

"P2200"; "T25000" -> the loop stopped oscillating

- By this approximation of the P share, a value can be found at which the loop starts to oscillate.

In our example, this happened at P = 2250 mA/K.

Read oscillation period at this value - for example, 5.58 sec.

The critical gain and the critical oscillation period are found.

4. Program the MTD415T with these values in order to trigger the calculation of the PID shares and the cycle time by the firmware:

Program the critical gain to the found above value: "G2250"

Program the critical oscillation period: "O5580"

5. The firmware calculates the parameters of the digital PID loop and applies them instantly. The calculated parameters can be read out using the following commands:

"P?" " [1350<LF>]" -> P share = 1350 mA/K

"I?" " [161<LF>]" -> I share = 161 mA/(K*s)

"D?" " [941<LF>]" -> D share = 941 (mA*s)/K

"C?" " [87<LF>]" -> Cycle time = 87 ms

6. Usually, at this point the PID optimization for the settling behavior is finished. If required, the PID values and the cycle time can be manually fine tuned in order to optimize the loop response to changes of the thermal load.
7. As a final step, save the settings to the non-volatile memory: "M"

Notes

- The cycling time is the time base of the internal digital control loop and is calculated automatically by entering G (critical gain) and O (critical oscillation period).
- If manually changing the cycling time ("C"), the firmware calculates the I and the D share anew.
- The optimized PID parameters are valid for a steady state, that depends on the set temperature as well as on the ambient conditions (ambient temperature, temperature of the thermally controlled object). Any changes of the operating and/or environmental conditions may require a re-adjustment of the PID parameters.

8 Troubleshooting

The MTD415T has an imprinted Data Matrix code, that contains manufacturing information:



Please scan this code with your smartphone, and submit the DataMatrix code information to [Thorlabs](https://www.thorlabs.com)²⁸.

Compatible Android apps are:



QR & Barcode Scanner **QRbot** by TeaCapps. This scanner allows to share the scanned code directly by email.

(<https://play.google.com/store/apps/details?id=net.qrbot&hl=de>)



QR Droid Code Scanner **QRDroid** by DroidLa.

(<https://play.google.com/store/apps/details?id=net.qrbot&hl=de>)

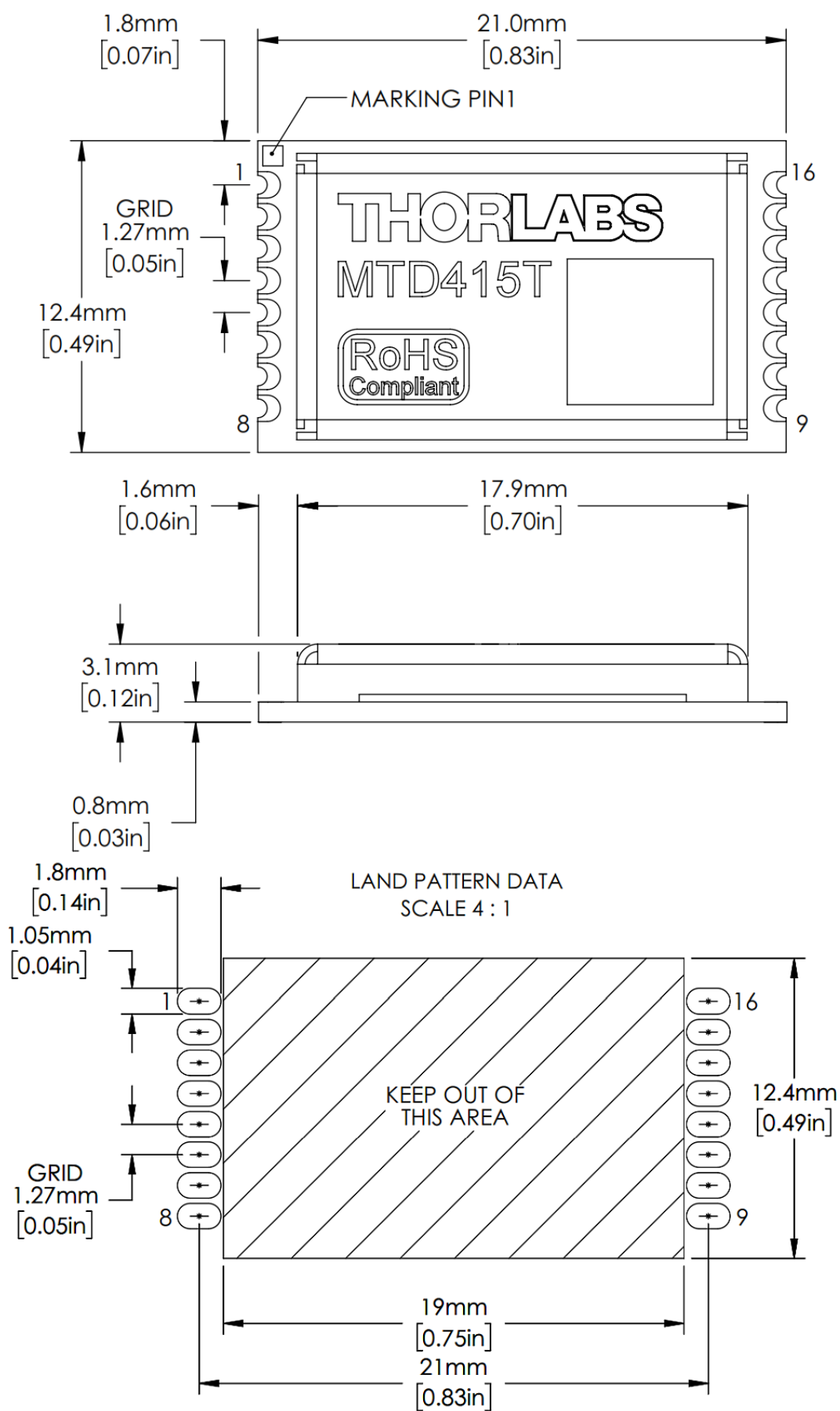
Compatible iOS apps:



QR & Mobile Barcode Scanner **NeoReader**® by NeoMedia Technologies.

(<https://itunes.apple.com/de/app/neoreader-gr-mobile-barcode/id284973754?mt=8>)

9 Drawing



10 List of Acronyms

The following acronyms and abbreviations are used in this manual:

AC	Alternating Current
DC	Direct Current
ESD	Electrostatic discharges
GND	Ground
GUI	Graphic User Interface
I ² C	Inter-Integrated Circuit
μC	Micro-Controller
NTC	Negative Temperature Coefficient (thermistor)
PCB	Printed Circuit Board
SCPI	Standard Commands for Programmable Instruments
SMT	Surface Mounted Technology
TEC	Thermoelectric cooler

11 Warranty

Thorlabs GmbH warrants material and production of the MTD415T for a period of 24 months starting with the date of shipment. During this warranty period Thorlabs GmbH will see to defaults by repair or by exchange if these are entitled to warranty.

For warranty repairs or service the unit must be sent back to Thorlabs GmbH. The customer will carry the shipping costs to Thorlabs GmbH, in case of warranty repairs Thorlabs GmbH will carry the shipping costs back to the customer.

If no warranty repair is applicable the customer also has to carry the costs for back shipment.

In case of shipment from outside EU duties, taxes etc. which should arise have to be carried by the customer.

Thorlabs GmbH warrants the hard- and/or software determined by Thorlabs GmbH for this unit to operate fault-free provided that they are handled according to our requirements. However, Thorlabs GmbH does not warrant a fault free and uninterrupted operation of the unit, of the software or firmware for special applications nor this instruction manual to be error free. Thorlabs GmbH is not liable for consequential damages.

Restriction of warranty

The warranty mentioned before does not cover errors and defects being the result of improper treatment, software or interface not supplied by us, modification, misuse or operation outside the defined ambient stated by us or unauthorized maintenance.

Further claims will not be consented to and will not be acknowledged. Thorlabs GmbH does explicitly not warrant the usability or the economical use for certain cases of application.

Thorlabs GmbH reserves the right to change this instruction manual or the technical data of the described unit at any time.

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This offer is valid for Thorlabs GmbH electrical and electronic equipment

- sold after August 13th 2005
- marked correspondingly with the crossed out “wheelie bin” logo (see figure below)
- sold to a company or institute within the EC
- currently owned by a company or institute within the EC
- still complete, not disassembled and not contaminated

As the WEEE directive applies to self contained operational electrical and electronic products, this “end of life” take back service does not refer to other Thorlabs GmbH products, such as

- pure OEM products, that means assemblies to be built into a unit by the user (e. g. OEM laser driver cards)
- components
- mechanics and optics
- left over parts of units disassembled by the user (PCB's, housings etc.).

Waste treatment on your own responsibility

If you do not return an “end of life” unit to Thorlabs GmbH, you must hand it to a company specialized in waste recovery. Do not dispose of the unit in a litter bin or at a public waste disposal site.

WEEE Number (Germany) : DE97581288

Ecological background

It is well known that waste treatment pollutes the environment by releasing toxic products during decomposition. The aim of the European RoHS Directive is to reduce the content of toxic substances in electronic products in the future.

The intent of the WEEE Directive is to enforce the recycling of WEEE. A controlled recycling of end-of-life products will thereby avoid negative impacts on the environment.



*Crossed out
"Wheelie Bin" symbol*

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